chain of over 100 domestic, private and small-scale enterprises that will provide employment to around 4,500 people.

According to HAL, two prototypes of the HTT-40 are under flight testing and it has demonstrated compliance with all major Preliminary Staff Qualitative Requirements (PSQR), including stall and spin (the corkscrew-like downward manoeuvre of an aircraft). "It successfully demonstrated six-turn spin capability on both right- and left-hand sides during flight tests. The HTT-40 can be flown continuously for six hours with an additional oxygen cylinder," says a top HAL official.

The first proposal for a new turboprop basic trainer was given in 1999. The HTT-40 was unveiled as a replacement for HAL's HPT-32 'Deepak' trainer, which the IAF had grounded in 2009 after it lost 19 pilots in multiple crashes.

That year, the MoD ruled that the IAF's requirement of 181 new trainers would be met through two simultaneous channels—75 aircraft would be imported while HAL would manufacture 106 HTT-40 trainers. The IAF vehemently opposed the indigenous project right away, citing high costs. Though the MoD initially backed the HTT-40 project, it eventually veered to the IAF's view that the proposed HTT-40 would be much costlier than the imported PC-7s. "There is no need for the HTT-40 trainer," Air Chief Marshal N.A.K. Browne (retd) had said at the 2013 Aero India show in Bengaluru. "We have the Pilatus PC-7, which is a proven aircraft. The HAL project is from scratch...the cost will be too high." The IAF had even asked HAL to scrap the HTT-40 and instead build PC-7 Mk IIs through tech transfer by Pilatus. In 2012, the IAF signed a deal for 75 Swiss aircraft for Rs 2,900 crore. In July 2013, Browne wrote to then defence minister A.K. Antony, insisting on the import of 106 more PC-7 trainers. Eventually, in 2019, the MoD decided to drop IAF's plans for more Pilatus PC-7s after the Central Bureau of Investigation accused the Swiss company of being involved in corrupt and fraudulent practices to bag the 2012 contract.

Meanwhile, with the HTT-40 project sidelined and the IAF choosing the Swiss trainer, T. Suvarna Raju, the then design head of HAL, approached its board for funds in 2014. His exertions bore fruit—approval was given to release Rs 400 crore from the company's funds to initiate the project. "I approached our board and told it that we were not getting government funds, so let's do this from HAL funds," Raju tells INDIA TODAY. He and his determined team had their goals set—even if the IAF did not buy it, the HTT-40 would be made available for exports, and even for flying schools.

Speaking on the condition of anonymity, a HAL functionary discloses that even as the designers were working on the HTT-40, other unforeseen obstacles nearly derailed it. He says that besides being starved of funds, a letter from a BJP member of Parliament to the chief vigilance commissioner, alleging corruption in engine selection for the HTT-40, delayed the process of acquisition of their engine of choice, which finally happened in July last year. "If we blame HAL for delays in delivering the trainer, we should not forget the hurdles it faced. From scuttling its budget to corruption allegations, HTT-40 developers went through all sorts of problems. It is the present regime that has made this possible under its Aatmanirbhar Bharat mission," says a key HAL official.

THE IAF CLAIMS IT NEVER OPPOSED THE HTT-40, BUT WAS WORRIED ABOUT ITS DELIVERY ON TIME BASED ON ITS IMMEDIATE NEEDS

Raju says that despite opposition from the IAF, the HTT-40 was designed in record time. Former defence minister Manohar Parrikar of the new NDA government backed the project. "Within 17 months (from 2015 to 2016)—from cutting the metal to flying the aircraft—I delivered it in record time," says Raju, who was the HAL chairman when the first prototype, bearing his initials 'TSR 001' on the wings, was introduced publicly in June 2016 in Bengaluru in Parrikar's presence.

One of the HTT-40's designers believes that the trainer has not only outperformed the Pilatus PC-7s but also exceeded the IAF's performance criteria. The IAF demanded a top speed of 400 kmph; the HTT-40 has been tested to 420 kmph, he points out. Test flights have also exceeded the IAF's ceiling requirement of 20,000 ft by 200 ft, he claims, adding that the aircraft can take off and land within just 800 metres of the runway, less than the 1,000-metre limit set by the IAF in the PSQR.

However, the IAF claims that it has never opposed the HTT-40 and that they were justifiably worried about its delivery on time vis-a-vis the immediate needs of the force. A top IAF official tells INDIA TODAY that while HAL opposed the import of the basic trainer aircraft in 2009 on the grounds that the HTT-40 was being developed, rarely is a new aircraft developed and certified in less than 10 years. The IAF, he insists, was only protesting against the Department of Defence Production of the MoD, HAL and the finance division of the MoD, who were trying to stop the import on the grounds that the HTT-40 would be ready soon.

"Imagine what would have happened to the IAF had it not procured the Pilatus PC-7s and waited for the HTT-40 since 2009. Even now, after signing the contract, the aircraft delivery would commence a few years from now," says the IAF official.

As the official points out, the IAF will get its full fleet of HTT-40s only around 2030. However, the Centre's decision to push for Indian military platforms can only be a step in the right direction. ■